Practice Sets: Student Instructions

Web Development Assessment: Backend API without a Database

Sets P1 (Product Catalog) and P2 (Library Books)

1. What this is

You get two practice papers, **P1** and **P2**. Each asks you to build a REST API from scratch, with the same style and rules as the assessment. Attempt both.

There is no evaluator for practice. You test your API by hand with Postman, Thunder Client or `curl`. Each paper has a checklist for this and explains how the real evaluation works.

2. What you need

- **Express** (Node.js 18 or newer) or **Spring Boot** (Maven and the JDK from class). You may use a different one for P2.
- **Node.js 18 or newer installed on your laptop, even if you use Spring Boot.** The assessment evaluator runs on Node.js. Check with `node --version`.
- An editor and an API testing tool.

3. Ground rules

1. Build from scratch. No starter code, no copied code.
2. No database, no files, no seed data. Records live in memory and the server starts **empty**.
3. No frontend.
4. Port **8080** (read `PORT`, default 8080).
5. Use the exact paths, field names and status codes in your paper. Do not rename anything.

4. Getting started

Express: run `npm init -y` and `npm install express`, create `server.js` using `process.env.PORT || 8080`, start with `node server.js`.

Spring Boot: generate a Maven project at start.spring.io with **Spring Web** only, put `server.port=${PORT:8080}` in `application.properties`, start with `mvn spring-boot:run`.

5. How to work through a paper

1. Read the whole paper. Sections 3 to 6 are the contract, section 7 has examples.
2. Build POST and GET by id first, then the list, PUT, PATCH and DELETE. Test each one.
3. Add pagination, filters, search and sort.
4. Check the “rules that are easy to miss” list, then run the checklist in section 9 of the paper.
5. **Restart your server** (so it is empty) and run the checklist again.

6. Mistakes that cost the most marks

- **Crashing on bad input.** Malformed JSON, a missing field or a wrong type must give `400` with a JSON body, never a crash or a web page.
- **Wrong status codes.** `201` create, `204` delete with no body, `404` unknown id, `409` duplicate.
- **A bare array as the list.** It must be an object with `data`, `page`, `limit`, `total`, `totalPages`.
- **Wrong `total`.** Count after filters and search, before pagination.

- **Numbers sorted as text.** 9 comes before `10`.
- **Case.** Search and text sort ignore case; enum values are exact.
- **Trusting the body.** Your server creates `id`; ignore any `id` sent.
- **A PUT that creates.** Unknown id is `404`.
- **Unknown routes.** They return a JSON `404`.

7. How you will be evaluated

On assessment day an evaluator creates its own random records and runs **100 checks**, 1 mark each. It checks only HTTP behaviour, not your code or framework. The assessment sets use different domains and field names, but the same rules.

Keep your practice projects. Your trainer may ask you to run and show them.

8. If you are stuck

Problem	What to do
Port already in use	An old server is still running. Stop it.
Request body arrives empty	Send `Content-Type: application/json` and enable JSON parsing.
HTML error page	Add a handler that returns JSON errors.
Unsure about a rule	Check the examples in section 7, then ask your trainer.

Good luck: **build, test, restart, test again.**